AURA 2K26

FIREGUARD

AUTONOMOUS FIRE-RESPONSE ROBOT CAR

PROJECT ABSTRACT

1. OUTLINE OF PROJECT IDEA

FIREGUARD is a compact autonomous robotic platform engineered for proactive fire safety and emergency response in indoor environments. Conventional fire safety systems rely primarily on alarm sounders to alert occupants, doing little to actively suppress localized flames before they escalate. FIREGUARD addresses this critical gap by integrating real-time flame detection, directional estimation, autonomous obstacle avoidance, and target approach with an automated water suppression unit into a single mobile platform.

By engaging fires in their incipient stage, the system minimizes structural damage, reduces smoke propagation risks, and prevents human operators from entering hazardous, toxic environments.

2. PROPOSED IMPLEMENTATION

The hardware architecture is driven by an **ESP32 microcontroller**, chosen for its dual-core processing capabilities, integrated Wi-Fi/Bluetooth modules, and ample General-Purpose Input/Output (GPIO) pins.

- **Flame Sensing:** Utilizes **three directional flame sensors** configured to detect infrared emissions from flames across Left, Center, and Right zones, allowing coarse directional mapping. A **temperature sensor** continuously monitors ambient environmental heat.
- **Mobility & Navigation:** Driven by **four DC geared motors** managed through a high-current motor driver circuit. For dynamic navigation, an **HC-SR04 ultrasonic sensor** mounted on an **SG90 servo motor** actively scans surroundings for obstacles across a 180-degree sweep.
- **Suppression & Display:** Water delivery uses an electronically controlled water pump with a targeted spray nozzle. Live operational status is rendered in real-time on an **OLED display**.

3. METHODOLOGY

The robot operates on a closed-loop autonomous execution sequence divided into three core operational phases:

- **Detection & Alignment:** Sensors scan continuously. When a flame is detected, the ESP32 evaluates sensor inputs to determine the fire's relative direction (Left, Center, or Right) and aligns the chassis accordingly.
- **Navigation & Avoidance:** The robot navigates toward the fire. If an obstacle blocks the path, the servo rotates the HC-SR04 sensor to scan clear angles, directing the motors to steer around the obstruction.
- **Suppression:** Upon reaching a designated safe operating distance from the flame, movement halts, a MOSFET switching circuit triggers the water pump, and water is sprayed through the nozzle to suppress the fire.





4. DESIRED OUTPUT & REAL-TIME OLED MONITORING

The primary output is a fully functional, self-navigating fire-suppression vehicle capable of autonomous localization and mitigation. During operation, the onboard OLED display visualizes live diagnostic telemetry for real-time monitoring:

Parameter	Status / Value	Parameter	Status / Value
FIRE DETECTED	YES	TEMPERATURE	42 °C
OBSTACLE	NO	WATER SPRAY	ON
DIRECTION	CENTER	SYSTEM	SUPPRESSING

5. APPLICATION SCOPE

FIREGUARD is tailored for immediate localized response in controlled indoor environments where early intervention prevents major hazards:

- Educational and university research laboratories
- Smart home environments and residential garages
- Small commercial offices and server rooms
- Institutional facilities and workshops

6. FUTURE SCOPE

Future improvements and scalability options for the FIREGUARD platform include:

- **Wireless & Remote Monitoring:** Integrating IoT modules for real-time remote telemetry logging and emergency alert notifications.
- **AI & Thermal Imaging:** Adopting computer vision, AI-based localization algorithms, and thermal imaging cameras for precision source targeting.
- **Advanced Navigation:** Implementing Simultaneous Localization and Mapping (SLAM) path-planning algorithms for complex environments.
- **Mechanical Enhancements:** Upgrading to automated multi-axis nozzle aiming, larger water capacity, and robust all-terrain mobility chassis.

7. SYSTEM BLOCK DIAGRAM

High-level data and hardware signal flow between sensing, processing, and actuation layers:

